

5G/6G Communication Systems- ECA5604 (LAB)

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Experiment 1: Simulation of FDMA in MATLAB

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

Objective 1:

Program:

```
clc; clear; close all;
```

```
fs=1000; t=0:1/fs:1;
```

```
TBW=600; Users=3;
```

```
CBW=TBW/Users;
```

```
fc=[100 300 500];
```

```
m=sin(2*pi*10*t);
```

```
s1=m.*cos(2*pi*fc(1)*t);
```

```
s2=m.*cos(2*pi*fc(2)*t);
```

```
s3=m.*cos(2*pi*fc(3)*t);
```

```
subplot(2,1,1)
```

```
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
```

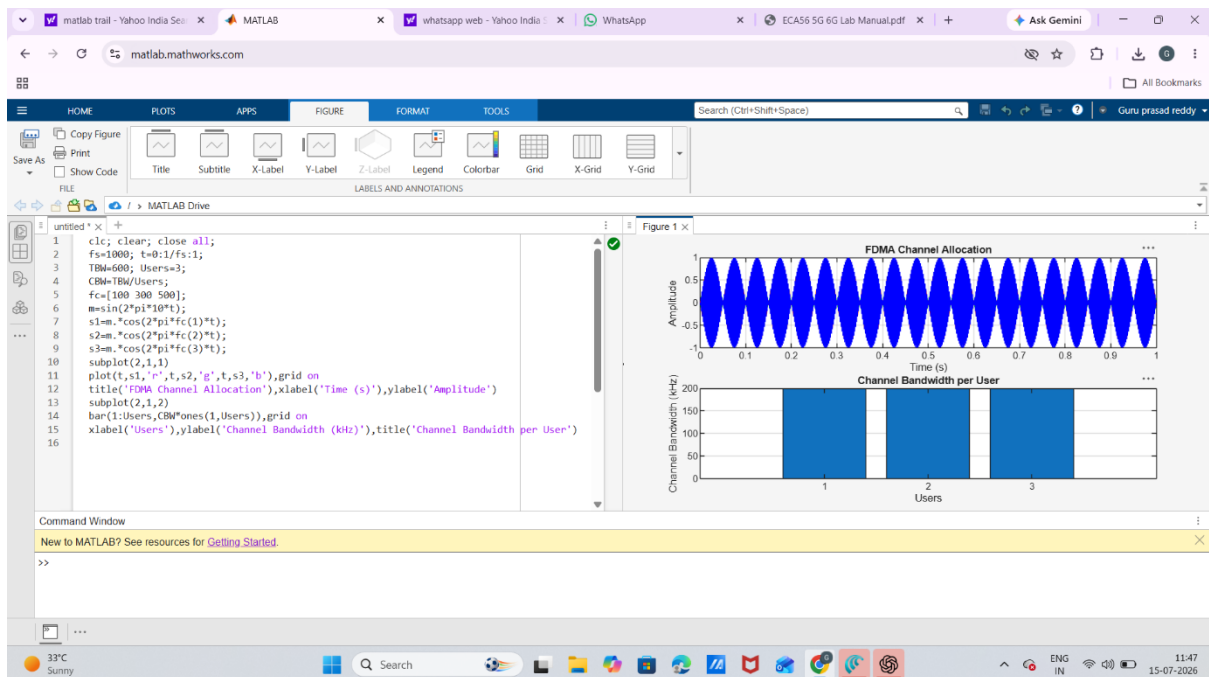
```
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
```

```
subplot(2,1,2)
```

```
bar(1:Users,CBW*ones(1,Users)),grid on
```

```
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

Output:



Objective 2:

Program:

clc; clear; close all;

BW = 200; % Channel Bandwidth (kHz)

Users = 3;

M = 2; % BPSK Modulation

DataRate = BW*log2(M); % Data Rate (kbps)

Rate = DataRate*ones(1,Users);

disp(' User Data Rate (kbps)')

disp([(1:Users)' Rate'])

figure

bar(1:Users,Rate)

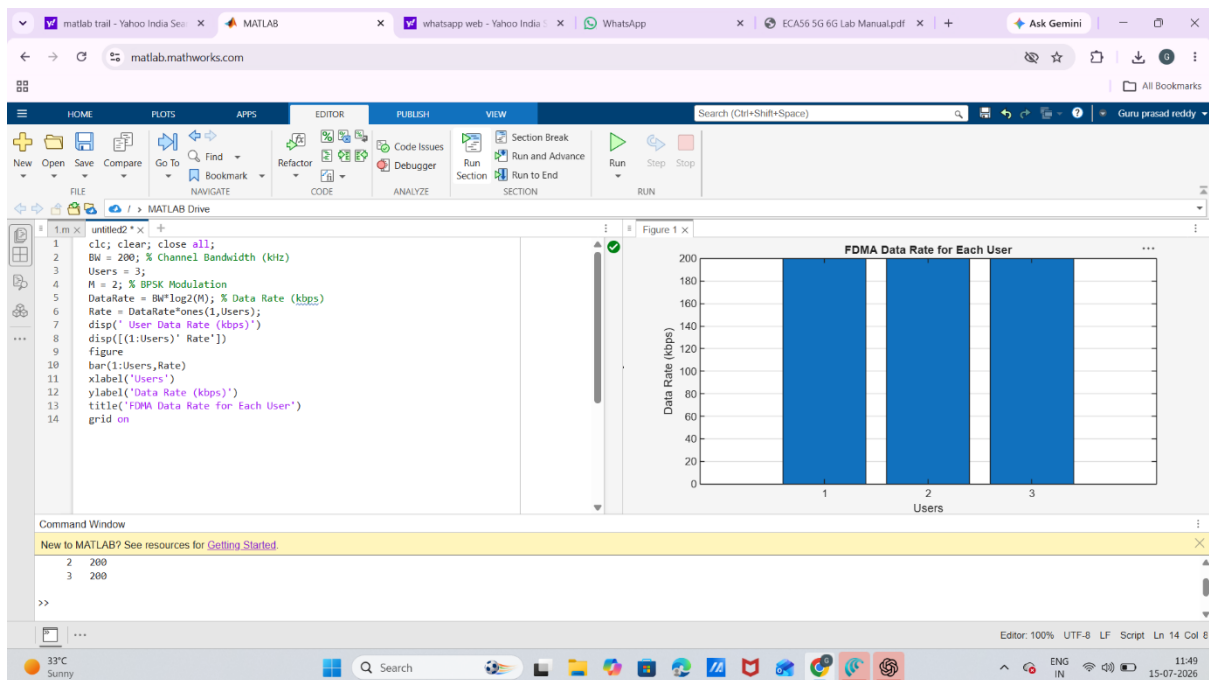
xlabel('Users')

ylabel('Data Rate (kbps)')

title('FDMA Data Rate for Each User')

grid on

Output:



Objective 3:

MATLAB Code:

```
clc; clear; close all;
```

```
Ps = 1; % Signal Power (W)
```

```
Pn = [0.1 0.05 0.01]; % Noise Power (W)
```

```
Users = 1:3;
```

```
SNR = 10*log10(Ps./Pn); % SNR in dB
```

```
disp('User Noise Power(W) SNR(dB)')
```

```
disp([Users' Pn' SNR'])
```

```
figure
```

```
bar(Users,SNR)
```

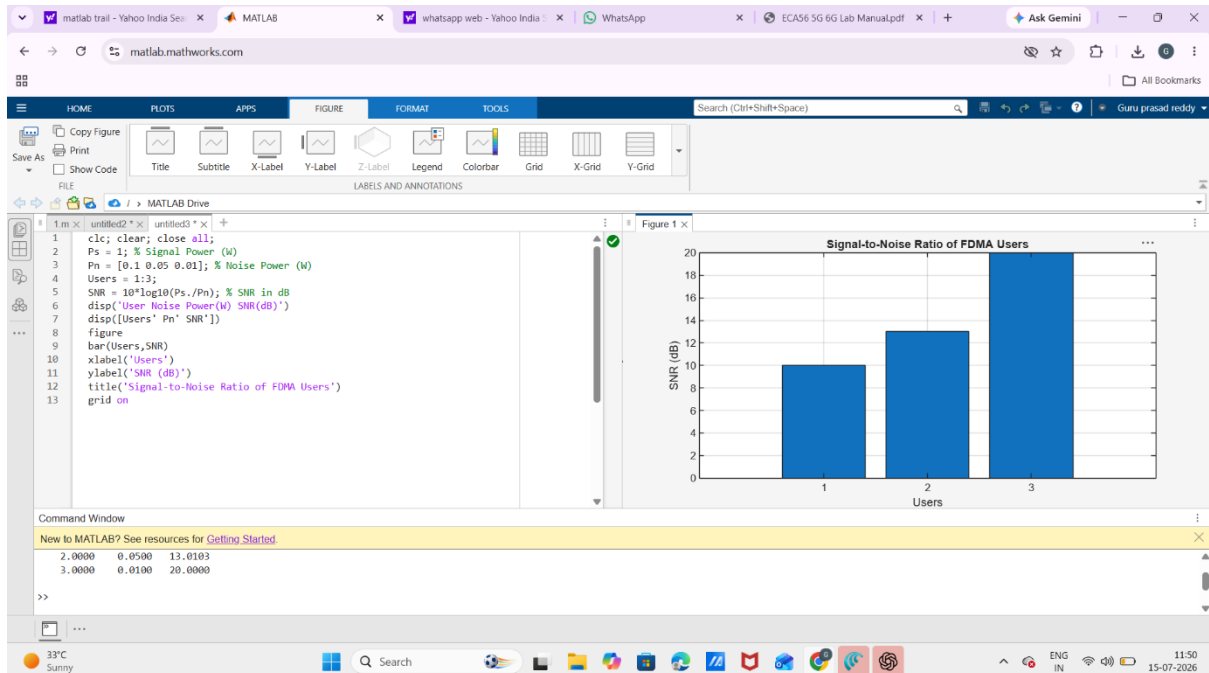
```
xlabel('Users')
```

```
ylabel('SNR (dB)')
```

```
title('Signal-to-Noise Ratio of FDMA Users')
```

```
grid on
```

Output:



Result:

The FDMA system was successfully simulated using MATLAB. The channel bandwidth, data rate, and Signal-to-Noise Ratio (SNR) for multiple users were calculated, demonstrating efficient frequency allocation and satisfactory communication performance.